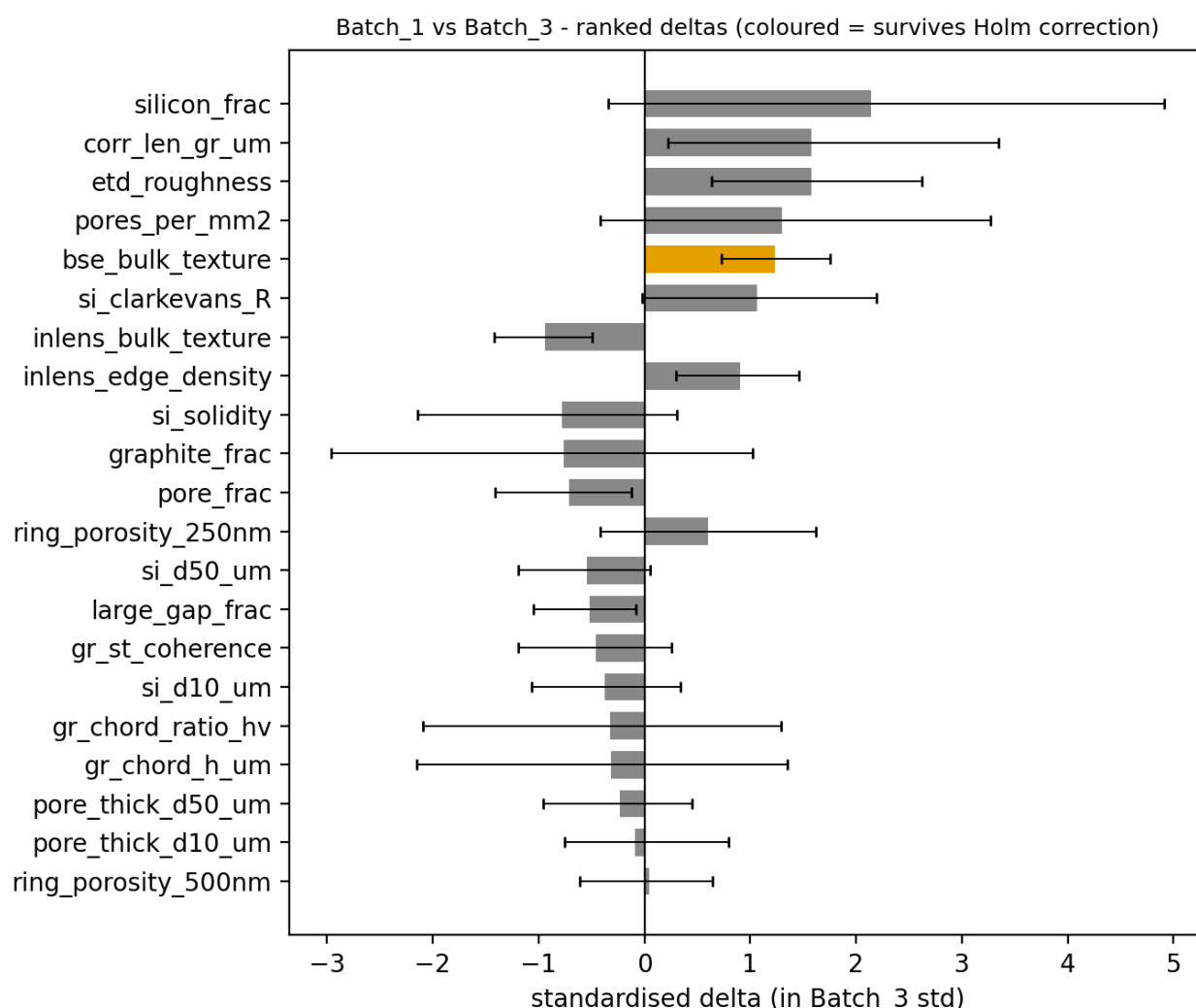
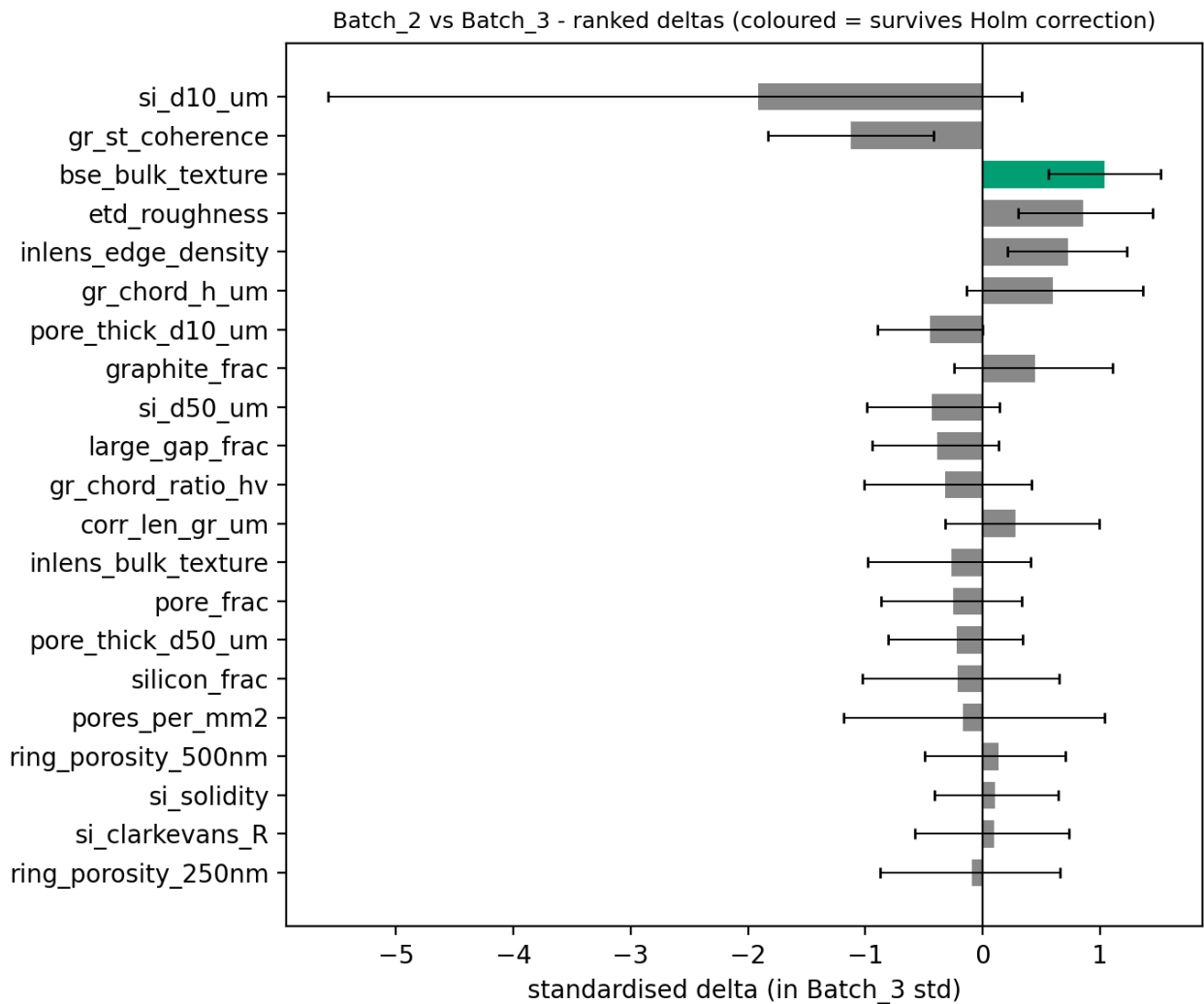


Phase 5 - Deltas vs Batch_3

Batch_1 vs Batch_3 (kept features): corr_len_gr_um +18% (suggestive); etd_roughness +18% (suggestive); bse_bulk_texture +18% (different, Holm ok); inlens_bulk_texture -21% (suggestive); inlens_edge_density +17% (suggestive); pore_frac -14% (suggestive); large_gap_frac -71% (suggestive). Batch_2 vs Batch_3: gr_st_coherence -8% (suggestive); bse_bulk_texture +15% (different, Holm ok); etd_roughness +10% (suggestive); inlens_edge_density +14% (suggestive); pore_thick_d10_um -7% (suggestive). With 7 images the smallest detectable standardised difference is ~1.0-1.5 B3 std (Mann-Whitney); intervals that do not survive Holm correction across the feature set are labelled 'suggestive', not proven.

Assumption (applies throughout): the grey bulk is assumed graphite, bright particles are assumed silicon-based, black is pore/crack. This comes from image appearance and is unconfirmed.





batch	feature	mean_batch	mean_b3	delta	pct_delta	std_delta	ci_lo	ci_hi	p_mw	p_holm	classification	mdd
Batch_1	silicon_frac	0.0969	0.06676	0.03014	45.15	2.146	-0.004834	0.06901	0.2878	1	undetermined	0.03692
Batch_1	corr_len_gr_um	1.1	0.9353	0.1647	17.61	1.58	0.02311	0.3488	0.07516	1	suggestive	0.1628
Batch_1	etd_roughness	0.03576	0.0303	0.005462	18.02	1.578	0.002185	0.009088	0.01602	0.3205	suggestive	0.003451
Batch_1	pores_per_mm2	1.718e+05	1.55e+05	1.684e+04	10.87	1.297	-5428	4.243e+04	0.4181	1	undetermined	2.393e+04
Batch_1	bse_bulk_texture	13.95	11.82	2.129	18.02	1.232	1.26	3.04	0.00026	0.005461	different	0.8903
Batch_1	si_clarkevans_R	0.7253	0.6793	0.04603	6.776	1.061	-0.0009891	0.09508	0.09949	1	undetermined	0.04803
Batch_1	inlens_bulk_texture	40.37	50.93	-10.56	-20.73	-0.939	-15.98	-5.575	0.03372	0.607	suggestive	5.205
Batch_1	inlens_edge_density	0.04693	0.04015	0.006778	16.88	0.9075	0.002201	0.01089	0.02826	0.537	suggestive	0.004344
Batch_1	si_solidity	0.8821	0.8959	-0.01382	-1.543	-0.7799	-0.03795	0.005486	0.3493	1	equivalent	0.02172
Batch_1	graphite_frac	0.8044	0.819	-0.01461	-1.784	-0.7594	-0.0569	0.01969	0.664	1	equivalent	0.0383
Batch_1	pore_frac	0.09871	0.1142	-0.01553	-13.6	-0.7144	-0.03063	-0.002615	0.1662	1	suggestive	0.01401
Batch_1	ring_porosity_250nm	0.09646	0.08355	0.01291	15.46	0.5984	-0.009024	0.03492	0.3493	1	undetermined	0.02197
Batch_1	si_d50_um	0.6522	0.763	-0.1108	-14.52	-0.5399	-0.2447	0.01112	0.3493	1	undetermined	0.1279
Batch_1	large_gap_frac	0.004253	0.01476	-0.01051	-71.19	-0.5205	-0.0212	-0.001589	0.1707	1	suggestive	0.009804

batch	feature	mean_batch	mean_b3	delta	pct_delta	std_delta	ci_lo	ci_hi	p_mw	p_holm	classification	mdd
Batch_1	gr_st_coherence	0.1047	0.1084	-0.00376	-3.467	-0.4632	-0.009699	0.002109	0.2337	1	equivalent	0.005904
Batch_1	si_d10_um	0.376	0.385	-0.008998	-2.337	-0.3749	-0.02566	0.008127	0.3493	1	equivalent	0.01689
Batch_1	gr_chord_ratio_hv	1.187	1.2	-0.01217	-1.014	-0.3216	-0.07922	0.04897	1	1	equivalent	0.06409
Batch_1	gr_chord_h_um	3.531	3.668	-0.1369	-3.731	-0.3198	-0.919	0.5765	0.7098	1	undetermined	0.7477
Batch_1	pore_thick_d50_um	0.259	0.2697	-0.0107	-3.967	-0.2326	-0.04375	0.02084	0.7748	1	undetermined	0.0323
Batch_1	pore_thick_d10_um	0.05296	0.05365	-0.0006961	-1.297	-0.08553	-0.006109	0.006439	1	1	undetermined	0.006274
Batch_1	ring_porosity_500nm	0.202	0.2	0.001978	0.9889	0.0486	-0.0249	0.02624	0.5761	1	undetermined	0.02557
Batch_2	si_d10_um	0.339	0.385	-0.04594	-11.93	-1.914	-0.1339	0.007998	0.7098	1	undetermined	0.07093
Batch_2	gr_st_coherence	0.09929	0.1084	-0.009142	-8.432	-1.126	-0.01488	-0.003411	0.01602	0.3205	suggestive	0.005734
Batch_2	bse_bulk_texture	13.61	11.82	1.794	15.18	1.038	0.9745	2.625	0.001046	0.02196	different	0.8251
Batch_2	etd_roughness	0.03328	0.0303	0.002975	9.817	0.8596	0.001041	0.005016	0.1471	1	suggestive	0.001987
Batch_2	inlens_edge_density	0.0456	0.04015	0.005445	13.56	0.7291	0.001598	0.009183	0.1662	1	suggestive	0.003793
Batch_2	gr_chord_h_um	3.925	3.668	0.2564	6.99	0.5992	-0.05764	0.5834	0.3828	1	undetermined	0.3205
Batch_2	pore_thick_d10_um	0.05	0.05365	-0.003655	-6.812	-0.4491	-0.007311	-6.336e-07	0.2919	1	suggestive	0.003655
Batch_2	graphite_frac	0.8276	0.819	0.008557	1.045	0.4449	-0.004731	0.0213	0.3828	1	equivalent	0.01302
Batch_2	si_d50_um	0.6748	0.763	-0.0882	-11.56	-0.4299	-0.2032	0.02912	0.5761	1	undetermined	0.1162
Batch_2	large_gap_frac	0.006959	0.01476	-0.007802	-52.85	-0.3865	-0.01895	0.00275	0.473	1	undetermined	0.01085
Batch_2	gr_chord_ratio_hv	1.187	1.2	-0.01214	-1.012	-0.3208	-0.03811	0.01588	0.5342	1	equivalent	0.02699
Batch_2	corr_len_gr_um	0.9643	0.9353	0.02899	3.099	0.2781	-0.03362	0.1038	0.6795	1	undetermined	0.0687
Batch_2	inlens_bulk_texture	47.9	50.93	-3.038	-5.964	-0.2701	-11.04	4.639	0.5342	1	undetermined	7.841
Batch_2	pore_frac	0.1087	0.1142	-0.005516	-4.828	-0.2537	-0.01882	0.007254	1	1	undetermined	0.01304
Batch_2	pore_thick_d50_um	0.2595	0.2697	-0.01015	-3.763	-0.2207	-0.03717	0.01578	0.924	1	undetermined	0.02648
Batch_2	silicon_frac	0.06372	0.06676	-0.003041	-4.556	-0.2166	-0.01439	0.009157	0.7098	1	undetermined	0.01177
Batch_2	pores_per_mm2	1.528e+05	1.55e+05	-2154	-1.39	-0.1659	-1.534e+04	1.353e+04	0.4939	1	equivalent	1.444e+04
Batch_2	ring_porosity_500nm	0.2053	0.2	0.00537	2.685	0.132	-0.02021	0.02882	0.3176	1	undetermined	0.02451
Batch_2	si_solidity	0.8978	0.8959	0.001818	0.2029	0.1026	-0.007334	0.01143	0.5761	1	equivalent	0.009382

Dropped/unreliable features (reported for completeness - do not drive conclusions):

batch	feature	mean_batch	mean_b3	delta	pct_delta	std_delta	ci_lo	ci_hi	p_mw	p_holm	classification	mdd
Batch_1	si_share_of_solid	0.1074	0.07523	0.03217	42.76	2.168	-0.006668	0.08025	0.3493	1	undetermined	0.04346
Batch_1	crack_len_density	0.1012	0.1283	-0.02703	-21.07	-1.685	-0.04734	-0.01013	0.005368	0.2845	suggestive	0.01861
Batch_1	si_per_mm2	7.501e+04	4.246e+04	3.256e+04	76.68	1.13	-1.289e+04	8.878e+04	0.4551	1	undetermined	5.084e+04
Batch_1	gr_sv	0.5846	0.6343	-0.04975	-7.843	-1.058	-0.08295	-0.02053	0.01602	0.8333	suggestive	0.03121
Batch_1	si_dmax_um	7.653	6.995	0.658	9.406	0.8637	-0.6071	1.97	0.4939	1	undetermined	1.289
Batch_1	crack_frac	0.04077	0.04698	-0.006208	-13.21	-0.7568	-0.01573	0.002394	0.2337	1	undetermined	0.009061
Batch_1	pore_largest_cc_frac	0.04414	0.0762	-0.03207	-42.08	-0.7377	-0.05591	-0.01104	0.06455	1	suggestive	0.02244

batch	feature	mean_batch	mean_b3	delta	pct_delta	std_delta	ci_lo	ci_hi	p_mw	p_holm	classification	mdd
batch_1	pore_depth_slope	-0.0008797	0.0208	-0.02168	-104.2	-0.5852	-0.04426	-0.001492	0.1139	1	suggestive	0.02138
batch_1	si_d50_area_um	3.298	3.058	0.2401	7.853	0.416	-0.1702	0.6589	0.4181	1	undetermined	0.4146
batch_1	gr_chord_v_um	2.933	3.054	-0.1212	-3.967	-0.3937	-0.6755	0.3708	0.8525	1	undetermined	0.5232
batch_1	pore_chord_h_um	0.6342	0.6695	-0.03529	-5.271	-0.371	-0.1162	0.03906	0.7565	1	undetermined	0.07764
batch_1	lineal_path_pore_um	0.6342	0.6695	-0.03529	-5.271	-0.371	-0.1094	0.03895	0.7565	1	undetermined	0.07415
batch_1	corr_len_si_um	1.657	1.565	0.09244	5.908	0.3496	-0.08865	0.2891	0.7508	1	undetermined	0.1889
batch_1	pore_chord_v_um	0.5388	0.5655	-0.02666	-4.715	-0.3367	-0.08563	0.03019	0.664	1	undetermined	0.05791
batch_1	pore_anisotropy	1.175	1.184	-0.00896	-0.7566	-0.2593	-0.05659	0.04052	0.8042	1	equivalent	0.04856
batch_1	pore_thick_d90_um	0.9018	0.9721	-0.07026	-7.228	-0.2559	-0.2457	0.1053	0.7268	1	undetermined	0.1755
batch_1	pore_patchiness_cv	0.1197	0.1099	0.009838	8.952	0.2449	-0.01514	0.03725	0.9014	1	undetermined	0.02619
batch_1	pore_depth_cv	0.1901	0.206	-0.01595	-7.742	-0.2253	-0.07257	0.04036	0.664	1	undetermined	0.05647
batch_1	si_no_pore_share	0.1787	0.1873	-0.008618	-4.602	-0.1743	-0.06055	0.04694	0.6194	1	undetermined	0.05374
batch_1	si_sv	2.261	2.164	0.09679	4.473	0.1656	-0.3736	0.6147	0.8042	1	undetermined	0.4942
batch_1	si_large_area_frac	0.7169	0.7018	0.0151	2.152	0.1633	-0.05116	0.08087	0.8525	1	undetermined	0.06602
batch_1	compact_pore_frac	0.05369	0.05251	0.001182	2.252	0.1468	-0.004107	0.006248	0.4939	1	undetermined	0.005177
batch_1	corr_len_pore_um	0.6357	0.6529	-0.01723	-2.638	-0.132	-0.1164	0.08403	0.924	1	undetermined	0.1002
batch_1	si_d90_um	2.308	2.367	-0.059	-2.492	-0.1121	-0.6637	0.4758	0.9506	1	undetermined	0.5698
batch_1	si_pore_dist_mean_um	0.9841	0.995	-0.01082	-1.088	-0.111	-0.07619	0.04996	0.9506	1	equivalent	0.06307
batch_1	si_patchiness_cv	0.2893	0.3044	-0.0151	-4.961	-0.109	-0.1239	0.09123	0.9506	1	undetermined	0.1076
batch_1	si_aspect	1.527	1.534	-0.006412	-0.4181	-0.08553	-0.06845	0.05397	0.9014	1	equivalent	0.06121
batch_1	si_d99_um	5.158	5.108	0.05017	0.9822	0.06614	-0.4883	0.5488	0.9506	1	undetermined	0.5186
batch_1	si_cracked_share	7.037e-05	0	7.037e-05	nan	nan	0	0.0002111	0.1425	1	suggestive	0.000105
batch_1	tau_x	inf	inf	nan	nan	nan	nan	nan	1	1	suggestive	nan
batch_1	tau_y	inf	inf	nan	nan	nan	nan	nan	1	1	suggestive	nan
batch_1	pore_percolates_x	0	0	0	nan	nan	0	0	1	1	suggestive	0
batch_1	pore_percolates_y	0	0	0	nan	nan	0	0	1	1	suggestive	0
batch_2	crack_len_density	0.1101	0.1283	-0.01817	-14.17	-1.133	-0.03099	-0.004313	0.02826	1	suggestive	0.01334
batch_2	compact_pore_frac	0.05959	0.05251	0.007089	13.5	0.8804	-0.001481	0.01546	0.1139	1	undetermined	0.008472
batch_2	gr_chord_v_um	3.306	3.054	0.2521	8.253	0.8191	0.01273	0.5092	0.1869	1	suggestive	0.2482
batch_2	gr_sv	0.6033	0.6343	-0.031	-4.887	-0.6591	-0.06807	0.006185	0.1471	1	undetermined	0.03713
batch_2	si_d50_area_um	3.434	3.058	0.3758	12.29	0.651	-0.0872	0.8067	0.1297	1	undetermined	0.4469
batch_2	si_patchiness_cv	0.3899	0.3044	0.08541	28.05	0.6162	-0.03326	0.2071	0.2094	1	undetermined	0.1202
batch_2	crack_frac	0.04218	0.04698	-0.004803	-10.22	-0.5855	-0.01066	0.0009706	0.1662	1	undetermined	0.005813